Developer Report: Terminal Task Tracker

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Purpose

The Terminal Task Tracker helps users manage their daily tasks through a command-line application. Users can add and edit active tasks, mark them as completed, and manage completed tasks by deleting them or moving them back to the active list.

Tools & Libraries

The application was built in Python 3 using VS Code and the standard library's `datetime` module to display the current date. The application uses separate Python modules to manage active tasks, completed tasks, and menu navigation, keeping the code organized without requiring external dependencies.

Challenges

The first version used recursive menu calls after each action and allowed users to attempt to complete tasks that did not exist. Reworking the menus to use loops and a shared submenu handler simplified navigation, while adding task validation prevented nonexistent tasks from being marked as completed.

Future Improvements

Tasks currently exist only in memory, so adding SQLite would allow them to persist between sessions. From there, I would add due dates, priorities, sorting, and filtering before expanding the application into a graphical or web-based interface.

Lessons Learned

This project reinforced the importance of planning before coding and how software engineering practices can improve quality and efficiency. Establishing the purpose and structure early made it easier to stay focused, address problems during development, and reduce unnecessary rework and technical debt.

How to Run the Application

1. Open the critical thinking project folder on GitHub.
2. Download the Python files into the same local folder.
3. Open a terminal in that folder and run `python main.py` (or `python3 main.py`, depending on your environment).